

NITIN NILESH

Bengaluru, India | +91-8981677732 | nitinnilesh49@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#) | [Google Scholar](#)
Applied Scientist | Machine Learning | LLM | Computer Vision & Camera Imaging | Graph ML

Applied Scientist with nearly 7 years of experience across production AI, video encoding and quality optimization, computer vision, graph ML, LLM/RAG applications, and imaging. Experienced in translating applied research from prototype to deployment, with strengths in model evaluation, perceptual quality metrics, optimization, agentic research, and business-impact measurement.

Core Skills

AI / ML	PyTorch, TensorFlow, Scikit-Learn, Graph ML, NumPy, Pandas, model evaluation, LoRA
LLM / GenAI	Transformers, RAG, LangChain, pgvector, CLIP, Segment Anything Model, Stable Diffusion, Multimodal Retrieval and Generation, Prompt Design (CoT, ReAct), Agentic AI workflows
MLOps / Inference	MLFlow, Weights & Biases, Docker, Containerization, ONNX, TensorRT, Mixed Precision, Quantization
Computer Vision	Video Understanding, Perceptual Image/Video Quality Assessment, Image Signal Processing, HEVC/X265, FFmpeg, OpenCV, CNN, VLM, ViT
Engineering	Python, C++, AWS, SQL, FastAPI, Git/GitHub, CI/CD Pipelines, Linux

Work Experience

Applied Scientist 2 - Amazon Prime Video | Bengaluru | Jan 2025 - Present

- Led design and production deployment of a **saliency-aware, mobile-optimized HEVC encoding ladder** for Amazon Prime Video mobile app, spanning **1080p to 144p** and reducing mobile streaming consumption to nearly **0.2 Gb/hour**.
- Designed a **CVBR optimization pipeline** around perceptual saliency and mobile constraints, achieving **~25% bitrate savings** while improving **floor VMAF** by **~20%** over the existing encode setup.
- Developed an in-house **ML-based Video Quality Metric** and evaluation framework scoring videos from 0-100, detecting artifacts such as banding, blocking, ghosting/motion blur, grain loss, luma shifts, and contrast degradation, and outperforming **Netflix VMAF** by **~6%** across SROCC, PLCC, and KROCC on subjective datasets.
- Built **LLM-powered agentic AI research workflows** for literature synthesis, hypothesis generation, experiment planning, execution analysis, and documentation, creating repeatable AI-assisted research pipelines for encoding-parameter optimization.

Generative AI Lead Engineer - Harman Connected Services | Bengaluru | Feb 2024 - Jan 2025

- Built and deployed a healthcare-domain **LLM/RAG chatbot** over hospital protocol documents using GPT, LangChain, pgvector, embeddings, and FastAPI; designed the end-to-end document-intelligence pipeline for ingestion, chunking, vector storage, semantic retrieval, and grounded response generation, converting static medical content into an interactive assistant that accelerated protocol discovery, reduced manual information lookup, and improved answer consistency and traceability.
- Developed **symptom-intake and summarization workflows** that made complex healthcare protocols easier to navigate for patients and enterprise teams while creating a reusable foundation for broader medical knowledge-assistant use cases.

Camera R&D Engineer - Qualcomm Research | Bengaluru | Sep 2021 - Nov 2023

- Developed, validated, and maintained an end-to-end **Qualcomm camera ISP pipeline** for raw Bayer image/video sensor data, covering black-level subtraction, lens correction, demosaicing, noise reduction, sharpening, white balance, color correction, tone mapping, and temporal filtering; enabled stage-wise defect isolation and intermediate-output analysis that shortened algorithm-improvement cycles, accelerated parameter tuning and integration, and reduced pre-silicon risk for premium-tier Qualcomm chipsets.
- Built a PyTorch-based **differentiable ISP research prototype** that optimized block-level algorithm parameters end to end using **SSIM loss**, producing perceptually tuned outputs and demonstrating a scalable path toward automated camera tuning, reduced manual tuning effort, faster sensor and use-case adaptation, and shorter camera-product development cycles.
- Worked with C++, Python, PyTorch, YUV data, and local compute infrastructure for camera pipeline validation, image-quality experiments, and ISP research.

Applied Scientist Intern - Amazon India Machine Learning | Bengaluru | Aug 2020 - Jan 2021

- Built large-scale heterogeneous customer and order graphs from 2 months of marketplace transactions, including **65.9M customer nodes** and **9.2M order-level nodes** across multiple identity and transaction edge types.
- Developed **Graph Convolution / Graph Attention models** with attention-based cross-edge-type aggregation for fraudulent customer and order classification, improving order-level **ROC-AUC from 0.90 to 0.95** and **PR-AUC from 0.111 to 0.179**.
- Demonstrated offline business impact through out-of-time inference, estimating **~\$3M in potential concession savings** by flagging 4,836 orders and 2,493 customers, translating large-scale graph ML results into actionable fraud-risk insights.

AI/ML Course Mentor - TalentSprint | Hyderabad | Sep 2018 - Dec 2021

- Mentored AI/ML courses in collaboration with IIIT-H Machine Learning Lab, creating tutorials, labs, and lectures for industry professionals.

Programmer Analyst - Cognizant Technology Solutions | Pune | Dec 2015 - Apr 2017

- Collaborated with the data science team to build machine-learning proof-of-concepts and strengthen early applied ML capability.

Selected Projects

Real-Time Structured Analysis for Broadcast Badminton Videos | Computer Vision, PyTorch

- Built a real-time system to extract structured analytics from live broadcast badminton videos, including player detection, localization, and on-court distance coverage during [Premier Badminton League 2019](#) [\[Website\]](#).

Neural Graph Execution | GNN, Optimization, PyTorch

- Created a pipeline for optimizing System-on-Chip workflows by using graph neural networks as an initial approximation before combinatorial optimization solvers.

Patents

- **System and Method for Implementing an Experiment Remotely and Determining an Output Using a Computer Vision Model** | U.S. Patent Application (18241852) | [USPTO](#) | Mar 2024
- **System and Method for Digitizing an Analog Water Meter Using Machine Learning** | Indian Patent Application (202141021341) | [Indian Patent Office](#) | Aug 2022

Selected Publications [\[Full List - Google Scholar\]](#)

- **AQIFormer: A Transformer-Based Multi-View Architecture for Cross-City Air Quality Classification** | 16th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) | [ACM](#) | 2025
- **TRAQID - Traffic-Related Air Quality Image Dataset** | 15th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) | [ACM](#) | 2024 | [\[Dataset\]](#)
- **Towards Real-Time Analysis of Broadcast Badminton Videos** | [arXiv preprint](#) | 2023 | [Code Blog](#)
- **IoT-based AQI Estimation using Image Processing and Learning Methods** | [WF-IoT](#) | 2022 | [Code Blog](#)

Education

MS by Research, Computer Science and Engineering | International Institute of Information Technology, Hyderabad | 2023

- Specialized in Computer Vision, Graph Neural Networks, NLP, and Large Language Models; advised by Prof. Sachin Chaudhari and Prof. C. V. Jawahar. [\[Thesis\]](#)

Bachelor of Technology, Computer Science and Engineering | Institute of Engineering and Management, Kolkata | 2015

Recognition and Courses

- Winner, [Environmental Sensing Project Competition 2022](#) organized by University of Helsinki for image-based AQI estimation technique.
- GATE 2017, CS: 98.39 percentile.
- Selected coursework: Deep Learning Specialization, Applied Machine Learning in Python, Programming/Data Structures/Algorithms using Python, Introduction to Machine Learning, Optimization Methods